

Same Orbit Implies Same Response: Strict Boundaries of Response Equivalence in Quantum Graph Heisenberg Models

Guanghao Li*

Independent Researcher, Hefei, Anhui, China

Rastislav Drahoš

Agora Research OS

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In quantum many-body systems, the response of a defect is often determined by its position. The standard answer is that two edges give the same response when they belong to the same edge automorphism orbit. We ask whether different orbits can also give the same response. We study the isotropic Heisenberg model on all connected graphs with $N = 6$ nodes (112 graphs) and $N = 7$ nodes (853 graphs). The fine-grained diagnostic is the standard deviation of all edge correlation functions, as a function of the coupling strength of a single defect edge. We find that same orbit implies same response with zero counterexamples, for both $N = 6$ and $N = 7$ (all 853 graphs, including the star graph and K_7). Different orbits can also give the same response: 47 pairs for $N = 6$ and 382 pairs for $N = 7$, not generated by involutive automorphisms. The mechanism is resolved as a layered structure. **Layer 1:** the total spin quantum number S of the two defect-edge states must be the same. This condition holds for all 47 pairs and excludes 274 of the 2731 different-orbit pairs, with zero false negatives. **Layer 2:** the absolute values of the Heisenberg coupling expectation at $s = 1/2$ must be equal. **Layer 3:** the sign rule is determined by S : for $S = 0$ the signs are equal, for $S = 1$ the signs are opposite. The sufficient condition is full $p_S(s)$ -curve equality, which selects 29 pairs (100% precision, 61.7% recall). The singlet probability admits the exact analytic form $p_S(s) = (1 - dE_0/ds)/4$ by the Hellmann-Feynman theorem; since $E_0(s)$ is concave, $p_S(s)$ is monotonically non-decreasing. Equal-value windows terminate exactly at the physical boundaries $s = 0$ (defect bond removed) and $s = 1$ (defect coupling equal to bulk). For $S = 1/2$ states, the Wigner-Eckart theorem gives $\langle \sigma_i \cdot \sigma_j \rangle = 3\langle \sigma_i^z \sigma_j^z \rangle$ exactly, so $p_S = 1/2$ implies $\langle \sigma \cdot \sigma \rangle = -1$ algebraically, not merely numerically. We also find 31 pairs of response-equivalent edges for $N = 6$ that exhibit gap exceptions, five of which take exact algebraic values -2 and $1 - \sqrt{3}$. Same orbit implies same response for all four interaction rules with zero counterexamples. All data are reproducible; independent verification scripts are available from the author.

I. INTRODUCTION

In quantum many-body physics, the position of a defect determines its response. Consider the Heisenberg model on a graph G . Choose one edge as the defect edge and vary its coupling strength s . The ground-state correlation distribution changes. When do two edges give the same response?

The standard answer is: when they belong to the same edge automorphism orbit. If a graph automorphism ϕ maps e_1 to e_2 , then the two Hamiltonians are conjugate under the permutation operator. The ground-state density matrix is invariant under the permutation. All edge correlation functions are pointwise identical. The sorted correlation distributions are identical. This is a direct consequence of graph automorphism symmetry.

But does this exhaust the truth? Standard physics assumes that the automorphism orbit is the complete classification. It does not ask whether different orbits can also give the same response. If they can, then there exists a coarser equivalence class. What is it? Is it char-

acterized by a known graph-theoretic quantity? Or does it require a new mathematical object?

We answer this question for the smallest non-trivial cases. We study all connected graphs with $N = 6$ nodes (112 graphs) and $N = 7$ nodes (853 graphs). We use exact diagonalization. We compute the fine-grained diagnostic for every edge as the defect edge. We compare all pairs of edges. We classify the pairs into four categories: same orbit same curve, same orbit different curve, different orbit same curve, different orbit different curve. We find that same orbit implies same response with zero counterexamples, for both $N = 6$ and $N = 7$.

For $N = 6$, to locate the mechanism, we scan 24 combinations of Hilbert space (full 2^N vs. $S_z = 0$), diagnostic (sorted correlation vector, standard deviation, range, entropy), and scan window ($[-1.5, 0.0]$, $[-3, 3]$, $[-4, 4]$). The union of different-orbit same-response pairs over the 24 combinations contains 47 pairs. The intersection contains 24 pairs.

The mechanism is resolved as a layered structure. **Layer 1:** the total spin quantum number S of the two defect-edge states must be the same. This condition holds for all 47 pairs and excludes 274 of the 2731 different-orbit pairs, with zero false negatives. **Layer 2:** the absolute values of the Heisenberg coupling ex-

* luoxuejian000@outlook.com

pectation at $s = 1/2$ must be equal. **Layer 3:** the sign rule is determined by S : for $S = 0$ the signs are equal ($sd_1 = +sd_2$), for $S = 1$ the signs are opposite ($sd_1 = -sd_2$). The sufficient condition is full $p_S(s)$ -curve equality, which selects 29 pairs (100% precision, 61.7% recall).

For non-degenerate ground states, the mechanism is resolved analytically. The Heisenberg ground state is an $SU(2)$ singlet, so the two-site reduced density matrix of the defect edge is always $SU(2)$ symmetric. We verify this at 4293/4293 sampling points with zero exceptions. The reduced density matrix is fully determined by a single scalar p_S , the singlet probability. By the Hellmann-Feynman theorem,

$$p_S(s) = \frac{1}{4} \left(1 - \frac{dE_0}{ds} \right). \quad (1)$$

Since $E_0(s)$ is concave, $p_S(s)$ is monotonically non-decreasing. Equal-value windows terminate exactly at $s = 0$ and $s = 1$.

For the core layer of 24 pairs, we test more than 20 candidate conditions. No single graph-theoretic condition exceeds 5% precision. The quantum condition $p_S(s)$ -curve equality is the strongest discriminator.

For degenerate ground states, the mechanism is more subtle. The $N = 6$ degenerate ground states have integer spin ($S = 0$ or $S = 1$), while the $N = 7$ ground states have half-integer spin ($S = 1/2$ with a long tail to $S = 2.5$).

The paper is organized as follows. Section II defines the model and observables. Section III describes the methods. Section IV reports the results. Section V discusses the relationship with standard physics and relational ontology. Section VI gives limitations and open problems. Section VII concludes. The appendices contain the analytical proofs.

II. MODEL AND DEFINITIONS

A. Hamiltonian

We use the isotropic Heisenberg model:

$$H = \sum_{(i,j) \in E} J_{ij} (\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z), \quad (2)$$

where the spin operators use the $S = \sigma$ convention. One edge, the defect edge, has coupling s . All other edges have coupling 1.0. We work in the $S_z = 0$ subspace. We use exact diagonalization. When the ground state is degenerate, we use the symmetry-reduced density matrix.

The choice $S = \sigma$ is a normalization. Earlier work with $S = \sigma/2$ gave gaps half as large and missed the values -2 and $1 - \sqrt{3}$. The physical conclusions are unchanged.

B. Fine-Grained Diagnostic

For a given s , the fine-grained diagnostic is

$$E(s) = \text{std} \left\{ \langle \sigma_i^z \sigma_j^z \rangle_s \right\}_{(i,j) \in E}. \quad (3)$$

This is the standard deviation of all edge correlation functions. It measures the dispersion of the correlation distribution. It is a relational observable: it depends on differences between edges, not on absolute values.

C. Sorted Correlation Distribution

At each s , we sort all edge correlation values. We obtain the sorted correlation distribution curve

$$\mathcal{R}_e(s) = \text{sort} \left\{ \langle \sigma_i^z \sigma_j^z \rangle_s^{(e)} \right\}_{(i,j) \in E}, \quad (4)$$

where the superscript (e) means edge e is the defect edge.

D. Response Equivalence

Two edges e_1 and e_2 are response-equivalent if and only if

$$\mathcal{R}_{e_1}(s) = \mathcal{R}_{e_2}(s) \quad (5)$$

for all s (tolerance 10^{-10}).

E. Two-Site Reduced Density Matrix and the Singlet Probability

For a defect edge $e = (i, j)$, we define the two-site reduced density matrix

$$\rho_{ij} = \text{Tr}_{\bar{i}\bar{j}} (\rho_{\text{GS}}), \quad (6)$$

where ρ_{GS} is the ground-state density matrix (symmetry-reduced if degenerate) and the trace is over all sites except i and j . The singlet probability is

$$p_S = \langle S | \rho_{ij} | S \rangle, \quad |S\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (7)$$

For an $SU(2)$ -symmetric ground state, the reduced density matrix takes the form

$$\rho_{ij} = p_S |S\rangle \langle S| + \frac{1-p_S}{3} (|T_+\rangle \langle T_+| + |T_0\rangle \langle T_0| + |T_-\rangle \langle T_-|), \quad (8)$$

with a single scalar p_S . The defect-edge correlation is

$$\langle \sigma_i^z \sigma_j^z \rangle = \frac{1}{3} - \frac{4p_S}{3}, \quad (9)$$

and the Heisenberg coupling expectation is

$$\langle \sigma_i \cdot \sigma_j \rangle = 1 - 4p_S. \quad (10)$$

This is the single-parameter characterization of the defect-edge state.

F. Analytic Form of $p_S(s)$

By the Hellmann-Feynman theorem applied to $H(s) = H_0 + s \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$,

$$\frac{dE_0}{ds} = \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle_{\text{defect}}. \quad (11)$$

Combined with $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma} \rangle = 1 - 4p_S$, this gives the exact analytic form

$$p_S(s) = \frac{1}{4} \left(1 - \frac{dE_0}{ds} \right). \quad (12)$$

Since $E_0(s)$ is the minimum of a family of functions linear in s , it is concave. Therefore $p_S(s)$ is monotonically non-decreasing. We verify this numerically: the derivative matches to 1.7×10^{-8} on 31,509 points, and p_S never decreases on any of the 10,503 defect edges.

For $S = 1/2$ states, the Wigner-Eckart theorem gives $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle = 3 \langle \sigma_i^z \sigma_j^z \rangle$ exactly. Hence $p_S = 1/2$ implies $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma} \rangle = -1$ algebraically, not merely numerically. The observed deviations from -1 at finite sampling are numerical: 206 of 417 misses are $S = 3/2$ or $5/2$ states where the Wigner-Eckart relation differs; 184 are Brent roots at level crossings; 27 are true roots missed by the tolerance on $\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}$.

G. Physical Boundaries of Equal-Value Windows

For a pair (e_1, e_2) , the maximal equal-value window over which $|sd_1(s)| = |sd_2(s)|$ terminates exactly at $s = 0$ (defect bond removed) or at $s = 1$ (defect coupling equal to bulk). At the termination point, at least one defect's ground-state total spin changes. The values -0.05 and 0.95 reported in earlier versions are the last grid points of a 0.05 grid before these two physical couplings; the exact boundaries are $s = 0$ and $s = 1$.

H. Involution Automorphism

Let ϕ be a node automorphism of G . ϕ is an involution if $\phi \circ \phi = \text{id}$. It is non-trivial if $\phi(v) \neq v$ for some v . An involution induces a pairing on edges: if $\phi(e_1) = e_2$ and $e_1 \neq e_2$, then e_1 and e_2 are a pair of involution images.

I. Ground-State Degeneracy

When the ground state is degenerate, a single eigenvector does not respect the automorphism symmetry. We use the symmetry-reduced density matrix:

$$\rho_{\text{GS}} = \frac{1}{d} \sum_{k=1}^d |\psi_k\rangle \langle \psi_k|, \quad (13)$$

where d is the degeneracy. Correlations are $\text{Tr}(\rho_{\text{GS}} \sigma_i^z \sigma_j^z)$.

For K_6 , the contrast is shown in Table I. A single eigenvector gives 105 same-orbit different-curve pairs. The symmetry-reduced density matrix gives zero. The explicit treatment of degeneracy is essential.

TABLE I. Degenerate ground state treatment for K_6 .

Treatment	Same orbit, same curve	Same orbit, different curve
Single eigenvector	0	105
Symmetry-reduced density matrix	105	0

J. Data Specifications

Tolerance: 10^{-10} . Degeneracy tolerance: 10^{-8} . Graph class: all connected graphs with $N = 6$ (112 graphs) and $N = 7$ (853 graphs). For $N = 7$, the scan interval is $s \in [-1.5, 0.0]$ with 11 points. For $N = 6$, we use three scan windows: $[-1.5, 0.0]$ (11 points), $[-3, 3]$ (61 points), and $[-4, 4]$ (241 points). For gap differences, step sensitivity uses 11, 21, 51, and 101 points. For the p_S curve analysis, we use a fine grid $s \in [-2.0, 2.0]$ with 81 points. The full $N = 7$ graph set includes the star graph (index 0) and K_7 (index 852), which have automorphism counts above 500.

III. METHODS

For each connected graph with $N = 6$ or $N = 7$:

1. Compute all node automorphisms. Obtain edge automorphism orbits and involution edge pairings.
2. For each edge as defect edge, scan s . Obtain the sorted correlation distribution curve.
3. For all edge pairs, compare curves. Compare whether they are involution images.
4. Classify into four categories.
5. Reverse scan: check whether all involution images have the same response.

We also record results under a single ground-state vector to expose the numerical trap.

For response-equivalent pairs, we compute correlation distribution entropy, maximum correlation, and gap. We test whether these are also equivalent.

For gap differences, we use four step sizes: 11, 21, 51, 101 points. We test whether the difference persists.

As a counterexample, we use seed 43. It is a small-world graph ($N = 8$, $K = 4$, $p = 0.1$) with no non-trivial involution. We verify that the number of $E(s)$ -identical edge pairs is zero.

a. *24-combination scan for $N = 6$.* To locate the mechanism of the different-orbit same-response pairs, we scan 24 combinations of the following three dimensions: Hilbert space (full 2^N or $S_z = 0$ subspace), diagnostic (sorted correlation vector, standard deviation, range, or Shannon entropy of the correlation distribution), and scan window ($[-1.5, 0.0]$, $[-3, 3]$, or $[-4, 4]$). For each combination we record which edge pairs are classified as different-orbit same-response. We then group the pairs by their 24-bit response pattern. The union over the 24 combinations contains 47 pairs; the intersection contains 24 pairs.

b. *Single-combination scan for $N = 7$.* For $N = 7$, the 24-combination scan is computationally heavier. We report the result from a single combination: full 2^N Hilbert space, sorted correlation vector, and scan window $[-1.5, 0.0]$ with 11 points. This gives 382 different-orbit same-response pairs.

c. *Two-site reduced density matrix computation.* To resolve the mechanism, we compute the two-site reduced density matrix ρ_{ij} for each defect edge on a fine grid $s \in [-2.0, 2.0]$ with 81 points. We extract the singlet probability p_S and the full set of Pauli correlations $\langle \sigma_i^a \sigma_j^a \rangle$ for $a = x, y, z$.

d. *Analytic form and monotonicity check.* We check the Hellmann-Feynman form $p_S(s) = (1 - dE_0/ds)/4$ by numerically differentiating $E_0(s)$ and comparing with the directly computed singlet probability. The derivative matches to 1.7×10^{-8} on 31,509 points. We verify that p_S is monotonically non-decreasing on all 10,503 defect edges.

e. *Layered mechanism search.* To resolve the mechanism of the 47 pairs, we scan all 2731 different-orbit pairs for $N = 6$ and test the following conditions:

1. **Layer 1:** total spin equality, $S_1 = S_2$, computed at $s = 1/2$ via the operator $S^2 = \sum_{ij} \sigma_i \cdot \sigma_j / 4$.
2. **Layer 2:** absolute coupling equality, $|\langle \sigma_i \cdot \sigma_j \rangle_1(s = 1/2)| = |\langle \sigma_i \cdot \sigma_j \rangle_2(s = 1/2)|$.
3. **Layer 3:** sign rule, $sd_1 = +sd_2$ for $S = 0$, $sd_1 = -sd_2$ for $S = 1$.
4. **Sufficient condition:** full $p_S(s)$ -curve equality.

For each condition, we compute precision and recall.

IV. RESULTS

A. Same Orbit Implies Same Response

After correct degeneracy treatment, the statistics for $N = 6$ (112 graphs) and $N = 7$ (853 graphs, including the star graph and K_7) are in Table II. Same-orbit different-curve is zero for both.

TABLE II. Exhaustive statistics for all connected graphs with $N = 6$ and $N = 7$. The $N = 7$ counts include the star graph (index 0) and K_7 (index 852), which have automorphism counts above 500; the corrected count is due to independent verification (see Acknowledgments).

Category	$N = 6$ (112 graphs)	$N = 7$ (853 graphs)
Same orbit, same curve	1066	6729
Same orbit, different curve	0	0
Different orbit, same curve	47	382
Different orbit, different curve	2684	44142

Same-orbit different-curve is zero for both $N = 6$ and $N = 7$. This is the hardest result of this paper.

B. Different Orbits Can Give the Same Response

The different-orbit same-curve pairs show that the equivalence class is coarser than the orbit. For $N = 6$, the union over 24 combinations contains 47 pairs. We tested candidate graph-theoretic quantities for $N = 6$. The results are in Table III. No quantity perfectly characterizes the class.

TABLE III. Candidate graph-theoretic quantities for $N = 6$.

Candidate invariant	Consistent graphs (out of 112)
Automorphism orbit	101
Adjacency spectral radius	98
Betweenness	68
Laplacian spectral gap	67
Endpoint degree pair	53
Degree sequence after deletion	53

C. Core Result: Involutive Automorphism Is Sufficient but Confined to Same-Orbit Pairs

1. Forward Direction

Let ϕ be a non-trivial involution, $\phi(e_1) = e_2$. Let $H_1(s)$ and $H_2(s)$ be the Hamiltonians with e_1 and e_2 as defect edges. Since ϕ is a graph automorphism and $\phi(e_1) = e_2$,

$$H_2(s) = U_\phi H_1(s) U_\phi^\dagger, \quad (14)$$

where U_ϕ is the permutation operator. The ground-state density matrix satisfies $\rho_2 = U_\phi \rho_1 U_\phi^\dagger$. For any edge f , the sorted correlation distribution is invariant under permutation. Hence $\mathcal{R}_{e_2}(s) = \mathcal{R}_{e_1}(s)$. This proves the forward direction.

Numerically, all involution images in all 112 graphs with $N = 6$ and all 853 graphs with $N = 7$ have the same response.

2. Reverse Direction Does Not Hold

On $N = 6$ and $N = 7$, all involution images lie within the same orbit. None of the different-orbit same-response pairs is an involution image.

D. Layered Mechanism for the 47 Pairs

The mechanism of the 47 different-orbit same-response pairs is resolved as a layered structure. We test the layers against all 2731 different-orbit pairs for $N = 6$.

1. Layer 1: Total Spin Equality is Universal

For each pair, we compute the total spin quantum number S_1 and S_2 of the two defect-edge states at $s = 1/2$, via $S^2 = \sum_{ij} \sigma_i \cdot \sigma_j / 4$. The 2×2 cross-table is in Table IV.

TABLE IV. Layer 1: total spin equality. $S_1 = S_2$ is satisfied by all 47 same-response pairs and by 2410 of the 2684 different-response pairs. All 274 pairs with $S_1 \neq S_2$ are different-response. Zero false negatives.

	Same response	Diff response	Total
$S_1 = S_2$	47	2410	2457
$S_1 \neq S_2$	0	274	274

All 47 same-response pairs satisfy $S_1 = S_2$. All 274 pairs with $S_1 \neq S_2$ are different-response. Zero false negatives. Layer 1 is a necessary condition.

The 47 pairs split into two branches: 40 pairs with $S = 0$, 7 pairs with $S = 1$.

2. Layer 2: Absolute Coupling Equality at $s = 1/2$

For each pair, we compute the Heisenberg coupling expectation at $s = 1/2$ and test whether $|sd_1| = |sd_2|$. The results are in Table V.

TABLE V. Layer 2: absolute coupling equality at $s = 1/2$. Recall is 100% for both S branches.

Branch	Pairs	Same resp	Recall	Precision
$S = 0$	2252	40	40/40 = 100%	40/110 = 36.4%
$S = 1$	205	7	7/20 = 100%	7/20 = 35.0%

Layer 2 recall is 100% for both branches. Layer 2 is a second necessary condition.

3. Layer 3: Sign Rule Determined by Total Spin

Given $|sd_1| = |sd_2|$, the sign rule is determined by the total spin S . For $S = 0$: $sd_1 = +sd_2$. For $S = 1$: $sd_1 = -sd_2$. The results are in Table VI.

TABLE VI. Layer 3: sign rule. $S = 0$ pairs satisfy $sd_1 = +sd_2$; $S = 1$ pairs satisfy $sd_1 = -sd_2$.

Branch	Sign rule	Recall
$S = 0$	$sd_1 = +sd_2$	40/40 = 100%
$S = 1$	$sd_1 = -sd_2$	7/7 = 100%

The sign rule is determined by the total spin quantum number.

4. Sufficient Condition: $p_S(s)$ -Curve Equality

The sufficient condition is full $p_S(s)$ -curve equality. The results are in Table VII.

TABLE VII. Sufficient condition: full $p_S(s)$ -curve equality. 29 pairs satisfy this condition, all of which are same-response pairs (100% precision, 61.7% recall).

Condition	Candidates	Same response
$p_S(s)$ -curve equality	29	29 (100%)
$sd(s)$ -curve equality	30	29 (96.7%)

The sufficient condition selects 29 pairs (100% precision, 61.7% recall).

5. Summary of Layered Mechanism

TABLE VIII. Layered mechanism. Layers 1–3 are necessary with zero false negatives; the sufficient condition selects 29 pairs.

Layer	Condition	Recall	Precision
1	$S_1 = S_2$	47/47 = 100%	47/2457 = 1.9%
2	$ sd_1 = sd_2 $ at $s = 1/2$	47/47 = 100%	47/130 = 36.2%
3	Sign rule: + for $S = 0$, – for $S = 1$	47/47 = 100%	47/47 = 100%
Sufficient	Full $p_S(s)$ -curve equality	29/47 = 61.7%	29/29 = 100%

E. Mechanism for Non-Degenerate Ground States: SU(2) Singlet-Probability Resonance

1. Step 1: SU(2) Symmetry and the Single-Parameter Theorem

The Heisenberg Hamiltonian is SU(2) symmetric. The ground state of a connected Heisenberg graph with even N is a total-spin singlet. We verify this numerically: for all 4293 sampling points across the 47 same-response pairs, the Pauli correlations satisfy

$$\langle \sigma_i^x \sigma_j^x \rangle = \langle \sigma_i^y \sigma_j^y \rangle = \langle \sigma_i^z \sigma_j^z \rangle, \quad (15)$$

with zero exceptions. The two-site reduced density matrix ρ_{ij} therefore takes the form

$$\rho_{ij} = p_S |S\rangle\langle S| + \frac{1 - p_S}{3} (|T_+\rangle\langle T_+| + |T_0\rangle\langle T_0| + |T_-\rangle\langle T_-|), \quad (16)$$

with a single scalar p_S .

2. Step 2: Exact Analytic Form of $p_S(s)$ and Monotonicity

By Hellmann-Feynman, $dE_0/ds = \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle_{\text{defect}}$, so

$$p_S(s) = \frac{1}{4} \left(1 - \frac{dE_0}{ds} \right). \quad (17)$$

Since $E_0(s)$ is concave, $p_S(s)$ is monotonically non-decreasing. See Appendix C for the derivation. Numerically, the derivative matches to 1.7×10^{-8} on 31,509 points, and p_S never decreases on any of the 10,503 defect edges.

3. Step 3: Resonance and Wigner-Eckart Identity

The resonance condition is an exact algebraic value of p_S :

$$p_S = 0 \iff \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle = +1, \quad (18)$$

$$p_S = \frac{1}{2} \iff \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle = -1, \quad (19)$$

$$p_S = 1 \iff \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle = -3. \quad (20)$$

For $S = 1/2$ states, the Wigner-Eckart theorem gives $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle = 3\langle \sigma_i^z \sigma_j^z \rangle$ exactly. Hence $p_S = 1/2 \iff \langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma} \rangle = -1$ holds algebraically, not merely numerically. In the 81-point scan, the value $+1/3$ appears at 1606/4293 = 37.41% with numerical precision 10^{-16} .

4. Step 4: Physical Boundaries of Equal-Value Windows

The maximal equal-value windows terminate exactly at $s = 0$ (defect bond removed) or $s = 1$ (defect coupling equal to bulk). For graphs 38, 60, and 103, at least one defect's ground-state total spin changes at the termination point. The values -0.05 and 0.95 reported in earlier versions are the last grid points of a 0.05 grid before these two couplings.

F. The Core Layer: Exhaustive Condition Search

TABLE IX. Exhaustive condition search for the core layer. No graph-theoretic condition exceeds 5% precision.

Condition	Candidates	Core hit	Precision
Graph-theoretic conditions			
share_endpoint	1488	13	0.9%
common_neighbors ≥ 4	2394	23	1.0%
both_not_bridge	2299	23	1.0%
contract_deg_equal	489	21	4.3%
combined_is_K4	146	4	2.7%
Quantum conditions			
p_S -curve equality	35	24	68.6%
p_S max diff $< 10^{-6}$	35	24	68.6%
p_S diff at $s = 0.5 < 10^{-6}$	113	24	21.2%

G. The 35 p_S -Equal Pairs: A Graph-Level Partition

TABLE X. The 35 p_S -equal pairs, partitioned by graph. No graph contributes to both core and middle.

Graph ID	Core pairs	Middle pairs	p_S mean
<i>Core-contributing graphs</i>			
9	1	0	0.789
26	1	0	0.227
33	4	0	0.516
49	4	0	0.522
55	4	0	0.522
64	1	0	0.167
69	9	0	0.049
<i>Middle-contributing graphs</i>			
31	0	1	0.173
42	0	1	0.077
83	0	4	0.177
106	0	2	0.069
108	0	3	0.060

TABLE XI. $p_S(s)$ -curve statistics for core and middle layers.

Layer	Pairs	$p_S@0$	$p_S@0.5$	$p_S@1$	p_S mean
Core	24	0.2007	0.3148	0.6118	0.2319
Middle	11	0.0818	0.1479	0.2775	0.1067

H. Three-Layer Structure

TABLE XII. Three-layer structure of the 47 different-orbit same-response pairs ($N = 6$).

Layer	Pairs	Pattern	p_S -equal	p_S regime
Core	24	all 24 bits	24/24	p_S curves identical on both edges
Middle	17	2-23 bits	11/17	depends on window/diagnostic
Outer	6	1 bit	0/6	single combination; graph 38

I. Mechanism for Degenerate Ground States: Numerical Observation

TABLE XIII. Degenerate -1 hit rate.

Quantity	$N = 6$	$N = 7$
Non-degenerate -1 hit rate	18/19	7486/7903
Degenerate -1 hit rate	19/29	561/1478
Degenerate spin type	integer ($S = 0, 1$)	half-integer ($S = 1/2, 3/2$)

J. Geometric Condition: Graph 38 and the Star-Triangle Structure

Graph 38 has the edge set $[(0, 1), (0, 2), (0, 3), (0, 4), (0, 5), (1, 2), (1, 3), (2, 3)]$: node 0 connects to all other nodes (star), and nodes 1, 2, 3 form a triangle. The degree sequence is $[1, 1, 3, 3, 3, 5]$, and the Laplacian spectrum is $[0, 1, 1, 4, 4, 6]$. Its equal-value windows terminate exactly at $s = 0$.

K. Response Equivalence Extends to Other Observables

TABLE XIV. Response equivalence extended to other observables for $N = 6$.

Observable	Equivalence rate
Correlation distribution entropy	1098/1098 = 100%
Maximum correlation value	1098/1098 = 100%
Gap	1069/1098 = 97.36%

L. Gap Exceptions: 31 Pairs for $N = 6$

TABLE XV. Structural analysis of 31 gap exceptions for $N = 6$.

Check item	Result
Ground-state energy curves identical	31/31
Ground-state degeneracy curves identical	31/31
Gap difference is constant	31/31
Difference occurs in first excited state	31/31
Step sensitivity test (4 step sizes)	31/31 persist

Among the 31 gap exceptions, five pairs take exact algebraic values. Graph 108 at $s = 0$ gives -2.0000000000 (precision 10^{-14}). Graph 109 at $s = 0.5$ gives $1 - \sqrt{3}$ (precision 10^{-14}).

M. Interaction Dependence

TABLE XVI. Response equivalence for four interaction rules ($N = 6$).

Rule	Same orbit, same	Same orbit, diff	Diff orbit, same	Diff orbit
Heisenberg	1066	0	32	2699
XXZ	1066	0	32	2699
Ising	1066	0	493	2238
XY	1066	0	23	2708

Same orbit implies same response for all four rules, with zero counterexamples.

N. Importance of Degeneracy Treatment

For K_6 , a single ground-state vector gives 105 same-orbit different-curve pairs. The symmetry-reduced density matrix gives zero.

V. DISCUSSION

A. Relationship with Standard Physics

The result “same orbit implies same response” is a direct consequence of graph automorphism symmetry. This work verifies it with zero counterexamples on all connected graphs with $N = 6$ and $N = 7$, for all four interaction rules, and delineates its boundary.

The mechanism is resolved as a layered structure. Layer 1 ($S_1 = S_2$) is universal. Layer 2 ($|sd_1| = |sd_2|$ at $s = 1/2$) holds for both S branches. Layer 3 (sign rule) is determined by the total spin. The sufficient condition is full $p_S(s)$ -curve equality. The singlet probability admits the exact analytic form $p_S(s) = (1 - dE_0/ds)/4$ and is monotonically non-decreasing. Equal-value windows terminate exactly at $s = 0$ or $s = 1$. This layered picture explains why no single graph-theoretic condition could discriminate the core layer: the mechanism is quantum, not graph-theoretic.

B. Relationship with Relational Ontology

Relational ontology states that the properties of an object are determined by its position in the relational network. Response equivalence is a concrete realization. The involution is a global symmetry of the relational field.

The different-orbit same-response pairs show that the relational position is not fully captured by the automorphism orbit. The mechanism is a layered quantum resonance: total spin equality, absolute coupling equality, sign rule, and p_S -curve equality.

VI. LIMITATIONS AND OPEN PROBLEMS

1. Finite size: exhaustive verification for $N = 6$ and $N = 7$. $N = 8$ requires full enumeration.
2. Layered mechanism beyond Layer 3: Layers 1, 2, 3 are necessary with zero false negatives. The sufficient condition (full p_S -curve equality) covers 29 of 47 pairs. The remaining 18 pairs require a weaker sufficient condition.
3. Core layer: no graph-theoretic quantity discriminates the 24 core pairs. The core and middle layers are distinguished by the p_S -curve height.
4. Degenerate ground states: for integer-spin ($N = 6$) and half-integer-spin ($N = 7$) degeneracies the -1 hit rate differs. A complete mechanism remains open.
5. Model dependence: Ising has 493 different-orbit same-response pairs, likely degeneracy artifacts.

VII. CONCLUSION

For all 112 connected graphs with $N = 6$ and all 853 connected graphs with $N = 7$ in the quantum Heisenberg model:

1. Same orbit implies same response: zero counterexamples for $N = 6$ and $N = 7$, and for all four interaction rules.
2. Different orbits can give the same response: 47 pairs for $N = 6$, 382 pairs for $N = 7$.
3. The mechanism for the 47 pairs is a layered structure. Layer 1: $S_1 = S_2$ (47/47 recall). Layer 2: $|sd_1| = |sd_2|$ at $s = 1/2$ (47/47 recall). Layer 3: sign rule determined by S (47/47 recall). The sufficient condition is full $p_S(s)$ -curve equality (29/29 precision, 29/47 recall).
4. The singlet probability admits the exact analytic form $p_S(s) = (1 - dE_0/ds)/4$ and is monotonically non-decreasing. Equal-value windows terminate exactly at the physical boundaries $s = 0$ and $s = 1$.
5. For $S = 1/2$ states, $p_S = 1/2 \Leftrightarrow \langle \sigma \cdot \sigma \rangle = -1$ holds exactly by the Wigner-Eckart theorem.
6. For the core layer of 24 pairs, no graph-theoretic quantity discriminates it. The core and middle layers are distinguished by the p_S -curve height.
7. For degenerate ground states, the mechanism remains open.

The boundary of the data is drawn. Same orbit implies same response. Different orbits may also give the same response, through a layered quantum resonance mechanism with analytic $p_S(s)$ and physical boundaries at $s = 0$ and $s = 1$.

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Appendix A: Analytical Proof of the Involutive Automorphism Theorem

Theorem: Let ϕ be a non-trivial involution of G , $\phi(e_1) = e_2$. Then the sorted correlation distribution with e_1 as defect edge is identical to that with e_2 as defect edge.

Proof: Since ϕ is a graph automorphism and $\phi(e_1) = e_2$, $H_2(s) = U_\phi H_1(s) U_\phi^\dagger$. The ground-state density matrix satisfies $\rho_2 = U_\phi \rho_1 U_\phi^\dagger$. For any edge f , $\langle \sigma_f^z \sigma_{f'}^z \rangle_2 = \langle \sigma_{\phi^{-1}(f)}^z \sigma_{\phi^{-1}(f')}^z \rangle_1$. The sorted distribution is invariant under permutation. Hence $\mathcal{R}_{e_2}(s) = \mathcal{R}_{e_1}(s)$. $\square$

Appendix B: Derivation of the SU(2) Singlet-Probability Theorem

Theorem: For an SU(2)-symmetric ground state, the two-site reduced density matrix ρ_{ij} is fully determined by a single scalar p_S , and the defect-edge correlation is

$$\langle \sigma_i^z \sigma_j^z \rangle = \frac{1}{3} - \frac{4p_S}{3}, \quad \langle \sigma_i \cdot \sigma_j \rangle = 1 - 4p_S. \quad (\text{B1})$$

Proof: The two-site Hilbert space decomposes into the singlet sector (dim 1) and the triplet sector (dim 3). SU(2) symmetry implies

$$\rho_{ij} = p_S |S\rangle\langle S| + \frac{1-p_S}{3} (|T_+\rangle\langle T_+| + |T_0\rangle\langle T_0| + |T_-\rangle\langle T_-|). \quad (\text{B2})$$

The z - z correlation is

$$\langle \sigma_i^z \sigma_j^z \rangle = p_S(-1) + \frac{1-p_S}{3}(+1-1+1) = \frac{1}{3} - \frac{4p_S}{3}. \quad (\text{B3})$$

By SU(2) symmetry, $\langle \sigma_i \cdot \sigma_j \rangle = 1 - 4p_S$. $\square$

This theorem is verified numerically at 4293/4293 sampling points with zero exceptions.

Appendix C: Derivation of the Analytic Form of $p_S(s)$

Theorem: For a defect edge with coupling s , the singlet probability satisfies

$$p_S(s) = \frac{1}{4} \left(1 - \frac{dE_0}{ds} \right), \quad (\text{C1})$$

and is monotonically non-decreasing.

Proof: Write $H(s) = H_0 + s \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$. By the Hellmann-Feynman theorem, $dE_0/ds = \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle_{\text{defect}}$. Combined with $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma} \rangle = 1 - 4p_S$, this gives the stated form. Since $E_0(s)$ is the minimum of a family of functions linear in s , it is concave; therefore dE_0/ds is non-increasing and $p_S(s)$ is non-decreasing. $\square$

Appendix D: Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Independent verification scripts and result files are available from the author upon reasonable request.

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